

# Resource Allocation and Optimization in Low-altitude D2D Communication Networks

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**Abstract**—In emergency scenarios, such as natural disasters where communication infrastructure is damaged, unmanned aerial vehicles (UAVs) can serve as aerial base stations, providing emergency coverage for internet of things (IoT) devices. This study adapts device-to-device (D2D) communication to the low-altitude environment, where UAVs act as flexible communication hubs, ensuring reliable data transmission in areas where traditional infrastructure is unavailable. The resource allocation challenges for D2D communication in low-altitude disaster response scenarios are addressed, where UAVs provide critical network coverage. To tackle these dynamic challenges, this paper proposes a Q-learning-based resource allocation method that operates in such complex environments. Under identical conditions compared with optimal methods, simulation results demonstrate that the proposed method improves energy and spectral efficiency, achieving approximately 11%–15% enhancement in system throughput. These results demonstrate the viability of UAVs as mobile base stations for expanding communication coverage in low-altitude economies, and they emphasize the importance of efficient resource management for D2D communication in such settings.

**Index Terms**—Low-altitude environment, Resource allocation, Reinforcement learning, Q-learning.

## I. INTRODUCTION

Over the past two decades, the rapid advancement of unmanned aerial vehicles (UAVs) technologies has significantly contributed to reshaping communication networks, especially in low-altitude airspace. As natural disasters increasingly disrupt terrestrial communication infrastructures, UAVs have emerged as pivotal technologies, providing mobile communication solutions in disaster-stricken areas. This trend is part of a broader shift towards the low-altitude economy, which leverages UAVs and urban air mobility (UAM) to enhance connectivity, especially in remote or damaged areas where traditional communication networks are unavailable or compromised. The demand for resilient communication networks is intensifying, particularly driven by the need for

emergency response, disaster management, and IoT applications in critical environments [1], [2], [3].

In this context, device-to-device (D2D) communication is crucial, as it allows UAVs and IoT devices to communicate directly without relying on traditional base stations. This technology is designed to extend coverage in disaster scenarios by utilizing the proximity of communication devices, thereby alleviating the burden on cellular networks and improving system throughput. D2D communication is especially beneficial in disaster-affected areas where traditional communication infrastructures are damaged, enabling rapid and reliable communication [4]. A distinctive strength of D2D communication lies in its seamless integration with other advanced technologies to further optimize wireless communication system performance [5]. This paper proposes a Q-learning algorithm for D2D communication resource allocation, leveraging reinforcement learning to quickly explore strategies by considering action spaces and their outcomes. It is proven effective in achieving higher throughput in complex low-altitude environments. The Q-learning framework ensures stable performance even with numerous users, making it ideal for dense urban deployments and large-scale implementations. Its dynamic adaptation to real-time channel conditions reduces interference, enhancing network stability, efficiency, scalability, latency, robustness, security, and mobility [6], [7].

Building upon constrained optimization theory, this study extends conventional resource allocation methodologies by proposing a Q-learning-based power allocation scheme. To validate algorithmic effectiveness, we construct two distinct communication scenarios incorporating obstacles and path loss characteristics, conducting comparative experiments against three benchmark power allocation approaches: random power allocation (Ran PA), path-loss-based power allocation (PL PA), and equal power allocation (Equ PA) [8].

This study focuses on achieving optimal resource allocation using Q-Learning under dynamic demand variations and diverse communication environments. The principal contributions are summarized as follows:

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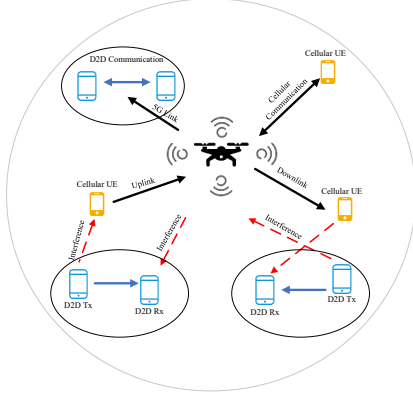


Fig. 1. System model for wireless cellular communication with D2D support.

The proposed study focuses on enhancing the efficiency of D2D communications in low-altitude environments, with a focus on dynamic resource allocation for UAVs acting as mobile base stations. By employing a Q-learning-based method, the study tackles the resource allocation challenges in this unique scenario, ensuring optimal network performance and resource usage in disaster response systems. The experimental design strictly adheres to standardized channel model parameter configurations to ensure result reproducibility.

## II. SYSTEM MODEL AND PROBLEM FORMULATION

### A. System Design

Cellular networks provide infrastructure support, including resource allocation, interference coordination, and QoS guarantee mechanisms, ensuring efficient coexistence between D2D links and traditional cellular services [9]. The system model is shown in Fig.1.

Two scenarios of cellular networks are considered in this paper. The first scenario (conventional underlaid cellular network with 5G protocol-occupied D2D pairs), a single-cell 5G cellular network with a radius of approximately 250 meters is simulated, where the base station (BS) is centrally located, and devices are randomly distributed within the cell. Mobile CUEs, D2D transmitters (D2D-Tx), and D2D receivers (D2D-Rx) are all assumed to reside within the BS coverage area. The configuration includes 30 D2D pairs and 20 CUEs for resource allocation. As illustrated in Fig.2, this scenario models a typical underlaid D2D network where 5G protocols govern spectrum sharing. The system aims to minimize energy consumption while prolonging terminal communication reliability.

The second scenario (mmWave band-based D2D communication in autonomous cellular networks). A constrained communication environment is simulated within a 200-meter radius single-cell autonomous network, where the BS is excluded from the cell due to impairment assumptions. Devices (CUEs, D2D-Tx, and D2D-Rx) are randomly distributed,

TABLE I  
SIMULATION PARAMETERS

| Parameter                                 | Value                  |
|-------------------------------------------|------------------------|
| Bandwidth, BW                             | 0.5GHz                 |
| Cell radius (First Scenario)              | 250m                   |
| Cell radius (Second Scenario)             | 200m                   |
| Number of CUE Devices                     | 20                     |
| Number of D2D Pairs                       | 30                     |
| Average blocking number                   | 35                     |
| Iterations                                | 50                     |
| Monte Carlo Repetitions                   | 1000                   |
| Total system energy                       | 35dBm                  |
| Pathloss parameter                        | 3                      |
| Minimum Power for D2D                     | $10^{(20/10^{-3})}$ W  |
| Maximum Power for D2D                     | $10^{(24/10^{-3})}$ W  |
| Total Power for D2D                       | $10^{(35/10^{-3})}$ W  |
| Total Power for Interferers               | $10^{(34/10^{-3})}$ W  |
| Frequency for Path Loss (First Scenario)  | 2GHz (microwave band)  |
| Frequency for Path Loss (Second Scenario) | 28GHz (microwave band) |
| Learning rate                             | 0.1                    |
| Exploration rate                          | 0.1                    |
| Discount factors                          | 0.9                    |
| Number of action levels                   | 5                      |
| Number of bins for status classification  | 5                      |
| Initial Q(S,A)                            | 0                      |
| Initial e(S,A)                            | 0                      |

with 30 D2D pairs and 20 CUEs allocated. Approximately 35 obstacles are spatially distributed following a poisson point process (PPP) to emulate real-world blockages. In this configuration, the system fully relies on D2D communication, while retaining limited functionality to facilitate initial device discovery and energy allocation. As shown in Fig.3, the BS operates at minimal capacity primarily assisting in device association and energy distribution with all subsequent operations delegated to devices.

This dual-scenario approach demonstrates the versatility of the QL PA method. It proves effective in both centrally-managed and fully ad-hoc contexts, covering a wide spectrum of real-world emergency and low-altitude communication use cases.

### B. Model Description

The proposed communication system model adopts a single-cell cellular architecture comprising a central BS and two categories of user equipment: M cellular user equipment (CUE) and N D2D communication pairs. These user sets are formally defined as  $CUE = \{C_1, C_2, \dots, C_M\}$  and  $D = \{D_1, D_2, \dots, D_N\}$ , where  $C_M$  and  $C_N$  denote the maximum supported numbers of cellular users and D2D pairs, respectively. Cellular users communicate via the BS while potentially serving as relay nodes in cooperative D2D transmission processes. All user devices (both cellular and D2D) are randomly distributed within the cell.

Each D2D pair consists of one transmitter (Tx) and one receiver (Rx). To maximize spectral efficiency, the system implements spectrum sharing between cellular and D2D users, with the number of available RBs equaling the cellular user count, i.e.,  $B = \{B_1, B_2, \dots, B_M\}$ . Here, a single RB is defined as the minimal allocation unit of spectrum resources.

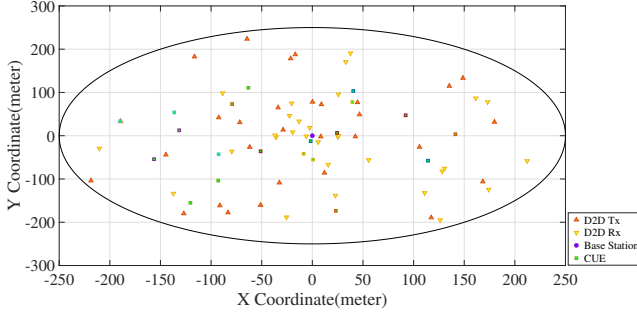


Fig. 2. Layout of the first scene.

Each D2D pair can occupy only one RB at any given time, while the same RB may be dynamically reused by multiple D2D pairs. Resource reuse introduces interference [10]. An interference threshold ITH is configured at an extremely low, negligible value while maintaining system QoS. The interference can be categorized as either cross-tier or co-tier. If both D2D connections and traditional communication links coexist between CUEs and the BS, this ultimately leads to inter-device interference. Specifically, D2D receivers experience interference from all active CUE-BS channels, Conversely, the BS is affected by in conventional cellular systems following this architecture, D2D receivers additionally suffer interference signals from other D2D pairs [10]. transmitted signals from D2D transmitters, Assume the  $i$ -th D2D pair is denoted as D2D( $i$ ), where  $i \in \{1, 2, \dots, k\}$ . The location of a D2D pair is given by two coordinate points( $q_1, p_1$ )and( $q_2, p_2$ ). The Euclidean distance formula provided in (1) is employed to calculate the distance between the  $i$ -th D2D pair

$$dist = \sqrt{(q_1 - q_2)^2 + (p_1 - p_2)^2}. \quad (1)$$

The study assumes an urban communication environment and applies a rayleigh fading path loss model expressed in dB, formulated as

$$PL_{D2D} = 12\log_{10}(dist) + 12\log_{10}(fc) + 19.45. \quad (2)$$

Path loss is typically expressed in dB units, while the channel gain  $h$  is a dimensionless ratio. To convert path loss into channel gain, the following formula is used

$$h = 10^{-\frac{PL_{D2D}}{10}}. \quad (3)$$

When attempting to determine channel quality and identify link throughput, computing SINR is crucial. The SINR formula for the first scenario is

$$\gamma_{K_k} = \frac{P_{D2D}(k) \cdot h_{D2D_{link}}(k)}{N_0 + I_{K_k}^{\mu W}}. \quad (4)$$

The SINR formula for the second scenario is

$$\gamma_{K_k} = \frac{P_{D2D}(k) \cdot h_{D2D_{link}}(k) \cdot G_{D2D}(k) \cdot PL_{D2D}(k)}{N_0 + I_{K_k}^{mmW}}, \quad (5)$$

where  $K \in \{Rx, BS\}$ ,  $k \in \{1, \dots, N\}$ ,  $P_{D2D}(k)$  denotes the transmit power of the  $k$ -th D2D pair,  $h_{D2D_{link}}(k)$  represents

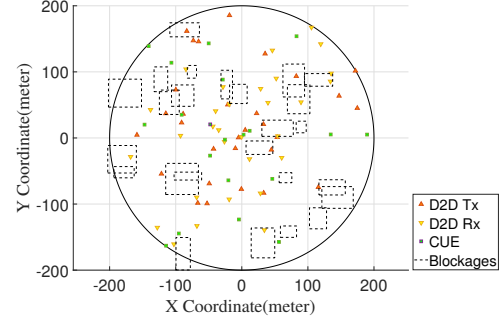


Fig. 3. Layout of the second scene.

the channel gain of the  $k$ -th D2D link,  $G_{D2D}(k)$  indicates the transmit antenna gain of the  $k$ -th D2D pair, and  $PL_{D2D}(k)$  is the path loss of the  $k$ -th D2D link.  $I_{K_k}^{\mu W}$  and  $I_{K_k}^{mmW}$  correspond to the interference power in the first and second scenarios respectively.

The noise power  $N_0$  is calculated by the following equation,

$$N_0 = -174 + 10\log(BW) + NF, \quad (6)$$

where  $BW$  denotes the bandwidth allocated for each transmission frequency, and  $NF$  represents the noise figure, a variable quantifying SNR degradation that is used in conjunction with radio receiver sensitivity.

The throughput calculation is generally based on Shannon's formula combined with SINR. The throughput for each link is expressed as

$$C = BW^{freq} \log_2(1 + \gamma_{K_k}), \quad (7)$$

$$T_{total} = \sum_{k=1}^N \cdot (BW^{freq} \log_2(1 + \gamma_{K_k})), \quad (8)$$

where  $freq \in \{\mu W, mmW\}$  denotes the frequency band parameter, and  $\gamma$  represents the SINR dependent on the receiver's scenario.

The first scenario adopts a conventional cellular network model. In the second scenario (millimeter wave band), path loss requires differentiation between LoS and NLoS states due to obstacle presence. For LoS links, the path loss equation can be expressed as a distance-dependent formulation,

$$PL_{LoS}(d) = \alpha_{LoS} + 10\beta_{LoS}\log_{10}(d) + \chi_{\sigma,LoS}. \quad (9)$$

In millimeter wave scenarios, generated obstacles may block communication links, resulting in NLoS states. The corresponding path loss formulation is defined as

$$PL_{NLoS}(d) = \alpha_{NLoS} + 10\beta_{NLoS}\log_{10}(d) + \chi_{\sigma,NLoS} + \Delta_{blockage}, \quad (10)$$

Where  $\alpha, \beta$  represent intercept and path loss index,  $x_{\sigma}$  represents shadow fading, and  $\Delta_{blockage}$  represents additional obstacle loss.

### III. REINFORCEMENT LEARNING STRATEGY FOR RESOURCE ALLOCATION

In dynamic network environments, traditional static resource allocation frameworks often lack adaptability, particularly when optimizing multi-dimensional metrics such as latency, energy efficiency, and fairness. To address these challenges, intelligent decision-making mechanisms optimized through environment-agent interactions have emerged as critical solutions. In real-time scenarios, distributed and dynamic resource allocation methods represent the most extensive and effective approaches [11]. When prior knowledge of systems, networks, users, and parameters is unavailable yet requires prediction and control decisions, machine learning techniques become applicable. Among these techniques, reinforcement learning (RL) monitors unknown parameters and system behaviors through trial-and-error learning and optimal control actions [12], [15]. As a model-free reinforcement learning algorithm, Q-learning has become particularly prominent in dynamic resource allocation due to its inherent ability to balance immediate rewards and long-term system objectives.

As shown in Fig.4, the experimental flowchart of Q-learning-based resource allocation for D2D communication demonstrates the following process: the system first initializes a Q-table containing multi-dimensional state-action pairs, where the state space consists of dynamic parameters such as real-time device location coordinates, channel quality indicator (CQI), and interference levels. The iterative process begins by observing the current state  $s_t$ , then selects an action  $a_t$  based on the observation results. After executing the chosen action, the environment provides feedback containing an immediate reward signal  $r_t$  and a new state observation  $s_{t+1}$ . Subsequently, the Q-learning algorithm updates the Q-value of the current state-action pair  $(s_t, a_t)$ , which incorporates both the obtained reward and expected future rewards. The Q-value update mechanism integrates current rewards with potential future gains, utilizing a learning rate parameter to control update speed while balancing short-term benefits and long-term network stability through a discount factor. This cycle repeats until algorithm convergence, enabling the D2D communication system to operate optimally in every environmental context. The ultimately converged Q-table provides D2D devices with an optimal resource mapping strategy, achieving adaptive spectrum efficiency optimization and cross-layer interference coordination in complex network environments [13], [14], [16].

Initialization of the Q-table: denoted as  $Q(s, a)$ , representing the expected cumulative reward for selecting action  $a$  in state  $s$ .

$$Q(s, a) \leftarrow 0 \cdots \text{or } Q(s, a) \sim \mathcal{U}(-\delta, +\delta). \quad (11)$$

The execution of an action  $a$  in (12) triggers a state transition  $s_t \rightarrow s_{t+1}$  and yields an immediate reward  $r$ .

$$r = \mathcal{R}(s_t, a, s_{t+1}), \dots, s_{t+1} \sim \mathcal{P}(s_{t+1} | s_t, a), \quad (12)$$

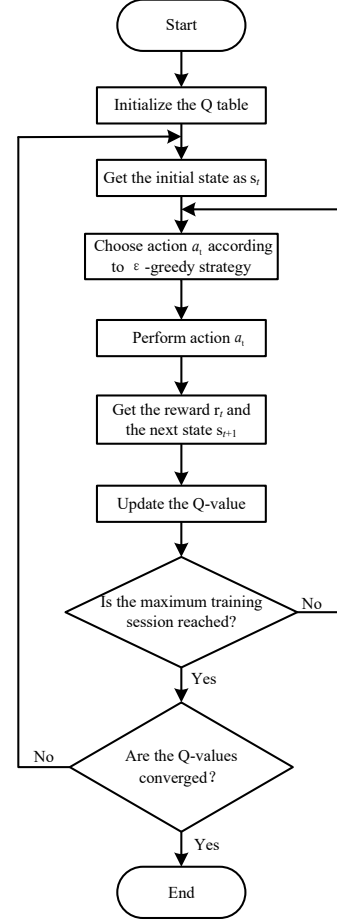


Fig. 4. Experimental flow chart of D2D communication resource allocation based on Q-learning.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma \max_{a'} Q_t(s_{t+1}, a_{t+1}) - Q_t(s_t, a_t)], \quad (13)$$

$$\delta = r_t + \gamma \max_{a'} Q_t(s_{t+1}, a_{t+1}) - Q_t(s_t, a_t), \quad (14)$$

$$r_t = \eta \cdot \log_2(1 + \gamma K_k) - \mu \cdot P_{\text{D2D}}(k), \quad (15)$$

Where  $\alpha \in (0, 1]$  denotes the learning rate, controlling the update magnitude of new information over previous values.

### IV. SIMULATION PARAMETERS AND RESULTS ANALYSIS

The proposed power distribution strategy is simulated using Matlab and numerically analyzed in this section. The simulation system parameters are summarized in Table I.

#### A. First Simulation Scenario

This section presents a quantitative analysis of simulation results for the first scenario. The simulation involved 50 training epochs, with each epoch comprising 450 iterations, resulting in a total of 22,500 iterative computations across all power allocation models. Fig.5 compares the system throughput of the QL PA strategy against the three benchmark power allocation methods (Equ PA, Ran PA, PL PA). To

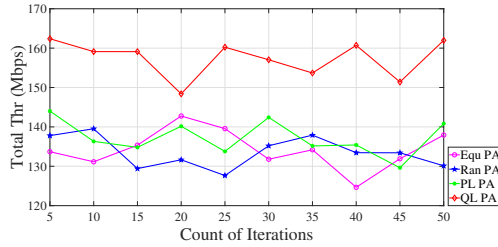


Fig. 5. Relationship between the number of iterations and system throughput in the first scenario.

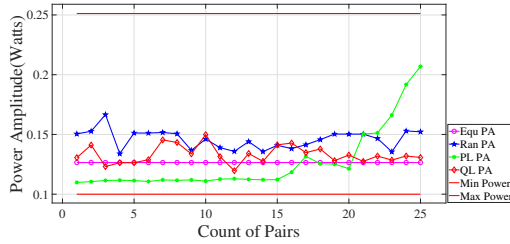


Fig. 6. Relationship between logarithm and power amplitude in the first scene.

mitigate transient fluctuations, output metrics were aggregated and averaged every 5 iterations. The results indicate that all four power allocation strategies exhibit oscillatory throughput patterns as iterations progress. Notably, the QL PA method demonstrates significantly enhanced throughput optimization compared to existing techniques. After 20 iterations, a quasi-periodic oscillation is formed, with an average of about 157.5Mbps, while the stable average of Equ PA, Ran PA, and PL PA is about 134.3Mbps, 133.6Mbps, and 137.2Mbps.

Fig.6 illustrates the correlation between the number of D2D pairs and power amplitude in the first scenario, depicting the average energy allocated per pair among 25 D2D pairs during simulations. The results demonstrate that the proposed QL PA closely follows Ran PA within the 7-17 pair range, with both strategies maintaining power levels within a relatively stable fluctuation range across this interval. Under constant total system energy constraints, Equ PA allocates identical energy per pair throughout. Notably, QL PA achieves higher overall power amplitude than Equ PA while exhibiting the second-lowest fluctuation magnitude after Equ PA. During later iterations, QL PA stabilizes power amplitude between 0.128W and 0.132W, ensuring signal reliability while optimizing energy conservation. In contrast, PL PA exhibits a monotonic increase in power amplitude with rising pair counts, with accelerated growth beyond 15 pairs.

The proposed QL PA achieves a peak data rate of 161.97Mbps during later iterations, outperforming PL PA (140.83Mbps) and Ran PA (130.12Mbps) by approximately 15% and 24%, respectively. In terms of power amplitude, QL PA operates within a constrained range of 0.126–0.131W across the 5–25 D2D pair interval, demonstrating smaller output fluctuations compared to PL PA's broader amplitude variations. While QL PA does not sustain maximum through-

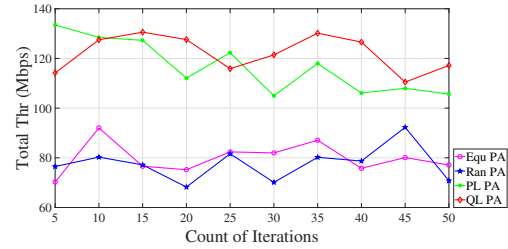


Fig. 7. Relationship between the number of iterations and system throughput in the second scenario.

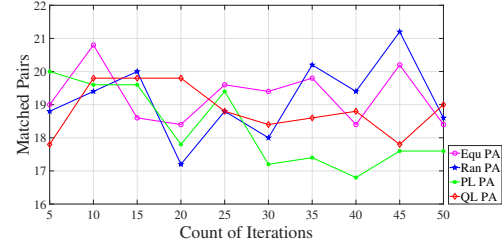


Fig. 8. Logarithmic relationship between the number of iterations and successful matching in the second scenario.

put all iterations, its progressive stabilization toward quasi-convergent behavior validates operational reliability under standard 5G cellular network conditions.

### B. Second Simulation Scenario

Fig.7 illustrates the relationship between iteration counts and system throughput in the second scenario. The simulation results show that the overall system throughput decreases by approximately 20.6–56Mbps compared to the first scenario. The stabilized mean throughput of QL PA is approximately 122.2Mbps, while those of Equ PA, Ran PA, and PL PA are approximately 79.8Mbps, 77.6Mbps, and 116.6Mbps, respectively. In the simulated scenario, when points are generated within obstacle areas, such points are removed from the available devices, thereby reducing the number of successfully linked pairs. Consequently, after creating the shortest-distance pairs, N is set to 25. Subsequently, if certain points are located within obstacles, the program removes both the point and its partner from the network. Fig.8 displays the instability of formed pairs during execution, with an average quantity of 18–21 pairs. The amplitude remains stable and outputs 0.187W at 16 pairs, while Equ PA, PL PA, and Ran PA exhibit transmit power levels of 0.168 W, 0.167W, and 0.157W, respectively. After a sharp decline, all four strategies show clear differences.

Fig.9 demonstrates the relationship between pair count and power amplitude during the first simulation execution in the second scenario. The simulation results indicate that QL PA maintains relatively stable power amplitude within the 1–16 pair range, achieving an output of 0.187W at 16 pairs. In contrast, Equ PA, PL PA, and Ran PA exhibit transmit power levels of 0.168W, 0.167W, and 0.157W, respectively. After a sharp decline, all four strategies showed clear differences. Notably, QL PA and PL PA exhibit noticeable enhancements



Fig. 9. Logarithm vs. power amplitude of the first simulation in the second scenario.

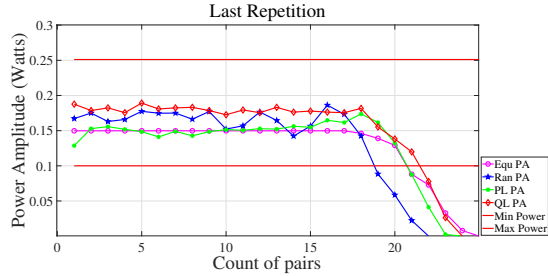


Fig. 10. Logarithm vs. power amplitude for the last simulation run in the second scenario.

prior to the abrupt declines at 15 and 16 pairs, indicating their adaptive power allocation responsiveness to complex network conditions.

Fig.10 illustrates the relationship between pair count and power amplitude during the final simulation execution in the second scenario. The simulation results demonstrate distinct behavior compared to the first execution: QL PA exhibits enhanced power amplitude stability within the 1–18 pair range. At 18 pairs, the output power levels for QL PA and the three comparative strategies are 0.182W, 0.146W, 0.174W, and 0.143W. All methods experience an abrupt decline beyond 18 pairs. Compared to alternative approaches, the proposed method proves capable of delivering optimal power allocation due to its superior energy efficiency metrics.

Despite the overall system throughput degradation, QL PA achieves a maximum data rate of 117.18Mbps in later iterations, outperforming PL PA (105.65Mbps) and Ran PA (70.78Mbps). Thus, for the considered D2D system model, the QL PA algorithm delivers approximately 11% higher output than PL PA and 66% higher output than Ran PA. Comparative analysis with the first scenario results reveals that QL PA is better suited for autonomous D2D communication in impaired transmission environments.

## V. CONCLUSION

This study proposes a Q-learning-based resource allocation framework tailored for D2D communications in low-altitude UAV-assisted networks, addressing challenges posed by dynamic and infrastructure-limited environments, such as those encountered during disaster response. By integrating adaptive learning and reducing communication overhead, this approach enables UAVs to act as mobile base stations for efficient D2D communication in both 5G and mmWave sce-

narios. Simulations show that the Q-learning-based strategy improves spectral and energy efficiency by 11%–15% over conventional methods, demonstrating strong environmental adaptability and real-time optimization potential for low-altitude economy applications. Future work will focus on three directions: multi-UAV collaborative resource allocation and trajectory planning, deep reinforcement learning (e.g., DQN) for scalability in complex settings, and energy- and privacy-aware designs such as federated learning.

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